

(12) **United States Patent**
Chung et al.

(10) **Patent No.:** **US 12,406,834 B2**
(45) **Date of Patent:** **Sep. 2, 2025**

(54) **SUBSTRATE PROCESSING APPARATUS AND SUBSTRATE PROCESSING METHOD**

(71) Applicant: **IUCF-HYU**
(INDUSTRY-UNIVERSITY COOPERATION FOUNDATION HANYANG UNIVERSITY), Seoul (KR)

(72) Inventors: **Chin Wook Chung**, Seoul (KR); **Moo Young Lee**, Seoul (KR)

(73) Assignee: **IUCF-HYU**
(INDUSTRY-UNIVERSITY COOPERATION FOUNDATION HANYANG UNIVERSITY), Seoul (KR)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 338 days.

(21) Appl. No.: **17/617,982**

(22) PCT Filed: **Apr. 1, 2020**

(86) PCT No.: **PCT/KR2020/004441**

§ 371 (c)(1),

(2) Date: **Dec. 10, 2021**

(87) PCT Pub. No.: **WO2020/251148**

PCT Pub. Date: **Dec. 17, 2020**

(65) **Prior Publication Data**

US 2022/0246407 A1 Aug. 4, 2022

(30) **Foreign Application Priority Data**

Jun. 11, 2019 (KR) 10-2019-0068789

(51) **Int. Cl.**

H01J 37/32 (2006.01)

C23C 16/50 (2006.01)

(52) **U.S. Cl.**

CPC **H01J 37/32697** (2013.01); **C23C 16/50** (2013.01); **H01J 37/32146** (2013.01); **H01J 37/3244** (2013.01); **H01J 2237/3341** (2013.01)

(58) **Field of Classification Search**

CPC H01J 37/32697; H01J 37/32146; H01J 37/3244; H01J 2237/3341; H01J 37/32;
(Continued)

(56) **References Cited**

U.S. PATENT DOCUMENTS

217,329 A * 7/1879 Briggs et al. B25B 1/08
269/130
4,659,449 A * 4/1987 Watanabe H01J 37/32431
156/345.39

(Continued)

FOREIGN PATENT DOCUMENTS

KR 10-2010-0012436 A 2/2010
KR 10-1010130 B1 1/2011

(Continued)

OTHER PUBLICATIONS

International Search Report of PCT/KR2020/004441 dated Jul. 15, 2020 [PCT/ISA/210].

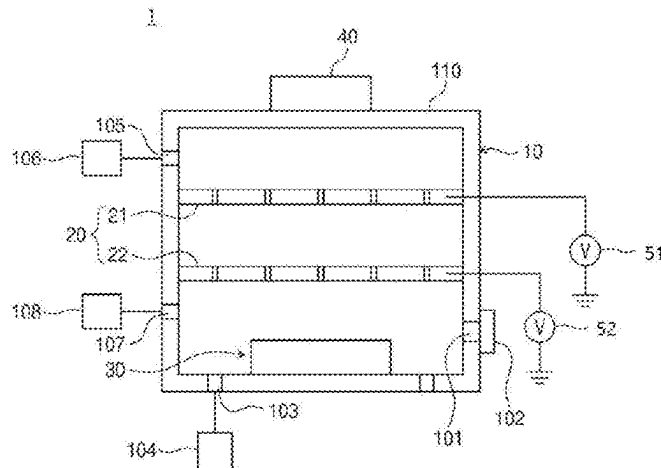
Primary Examiner — Rudy Zervigon

(74) *Attorney, Agent, or Firm* — Sughrue Mion, PLLC

(57) **ABSTRACT**

The present invention relates to a substrate processing apparatus and a substrate processing method. A substrate processing apparatus according to an embodiment of the present invention includes a chamber; a susceptor located inside the chamber to support a substrate to be processed; a first plasma control plate located in an upper area of a space inside the chamber; a second plasma control plate located below the first plasma control plate while being spaced apart from the first plasma control plate by a predetermined distance; and a plasma control power source connected to

(Continued)



the first plasma control plate and the second plasma control plate.

5 Claims, 3 Drawing Sheets

(58) Field of Classification Search

CPC H01J 37/32174; H01J 37/32935; H01J 37/3299; H01J 37/32972; H01J 37/32963; C23C 16/50; C23C 16/45565; C23C 16/45536; C23C 16/52; C23C 14/54; H01L 21/02; H01L 21/67; H01L 21/02274; H01L 21/67017; H01L 21/67253; H01L 21/67248; H01L 22/20
USPC 118/723 ER, 723 E, 663, 665, 668, 688; 156/345.45, 345.24, 345.25, 345.26, 156/345.27, 345.28

See application file for complete search history.

(56) References Cited

U.S. PATENT DOCUMENTS

5,154,943 A * 10/1992 Etzkorn C23C 16/455 427/166
6,511,575 B1 * 1/2003 Shindo B08B 7/0035 257/E21.582
6,656,540 B2 * 12/2003 Sakamoto C23C 16/452 156/345.46
7,244,474 B2 * 7/2007 Hanawa C23C 16/452 427/574
7,329,608 B2 * 2/2008 Babayan H01L 23/08 427/535
7,695,590 B2 * 4/2010 Hanawa C23C 16/045 156/345.43
8,608,853 B2 * 12/2013 Tseng H01L 21/67098 261/22
9,144,147 B2 * 9/2015 Yang H01J 37/32449
10,256,079 B2 * 4/2019 Lubomirsky C23C 16/505
10,403,507 B2 * 9/2019 Choi H01J 37/32357
10,697,062 B2 * 6/2020 Kao C23C 16/45587
10,727,080 B2 * 7/2020 Wang H01L 21/67069
10,903,054 B2 * 1/2021 Singh H01L 21/02049
11,062,920 B2 * 7/2021 Berry, III H01L 21/3065
2002/0129902 A1 * 9/2002 Babayan H01J 37/32009 156/345.43

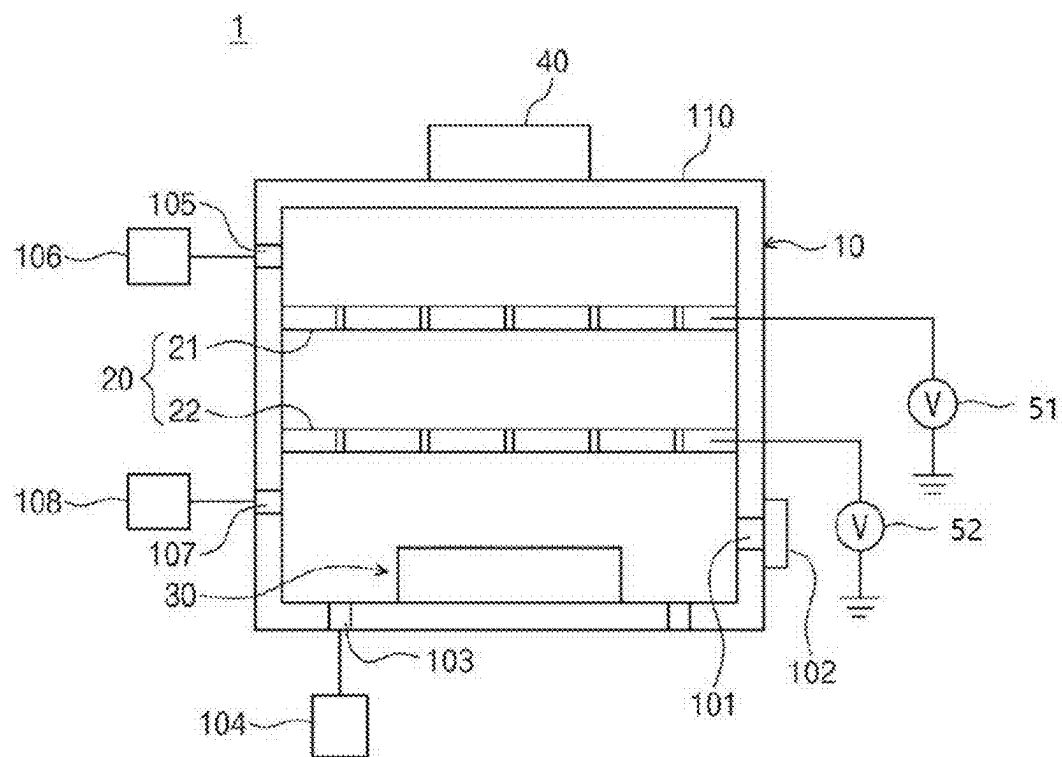
2003/0062840 A1 4/2003 Moroz
2004/0144490 A1 * 7/2004 Zhao H01J 37/32082 156/345.47
2004/0194890 A1 * 10/2004 Moroz H01J 37/32082 156/345.48
2004/0219737 A1 * 11/2004 Quon H01J 37/32697 438/222
2005/0211170 A1 * 9/2005 Hanawa H01J 37/32449 118/723 R
2005/0211171 A1 * 9/2005 Hanawa C23C 16/507 118/723 R
2005/0211546 A1 * 9/2005 Hanawa H01J 37/32449 204/192.12
2005/0211547 A1 * 9/2005 Hanawa C23C 14/0036 204/192.12
2005/0214477 A1 * 9/2005 Hanawa C23C 16/452 427/569
2005/0214478 A1 * 9/2005 Hanawa C23C 16/452 427/569
2006/0019039 A1 * 1/2006 Hanawa C23C 16/045 427/523
2006/0019477 A1 * 1/2006 Hanawa C23C 14/48 257/E21.279
2014/0273411 A1 * 9/2014 Abedijaberi H01L 21/02381 438/478
2014/0283746 A1 * 9/2014 Seo C23C 16/513 118/723 R
2014/0284404 A1 * 9/2014 Kuah C23C 16/45565 239/408
2015/0170943 A1 * 6/2015 Nguyen H01J 37/32082 156/345.33
2016/0217979 A1 * 7/2016 Kim H01J 37/3244
2016/0289830 A1 * 10/2016 Abedijaberi C23C 16/52
2017/0040170 A1 * 2/2017 Guha H01L 21/67069
2021/0074558 A1 * 3/2021 Nam H01J 37/32715
2022/0165546 A1 * 5/2022 Lill H01J 37/32082
2022/0307138 A1 * 9/2022 Kordina B01F 25/4336
2023/0313374 A1 * 10/2023 Kim C23C 16/45563 118/715

FOREIGN PATENT DOCUMENTS

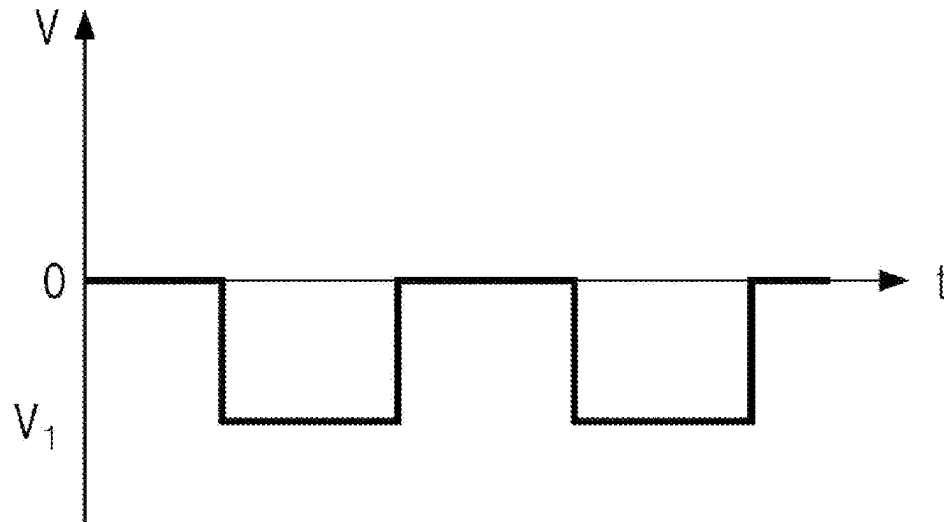
KR 10-2013-0005840 A 1/2013
KR 10-2015-0125829 A 11/2015
KR 10-1607520 B1 3/2016
KR 10-1717629 B1 3/2017

* cited by examiner

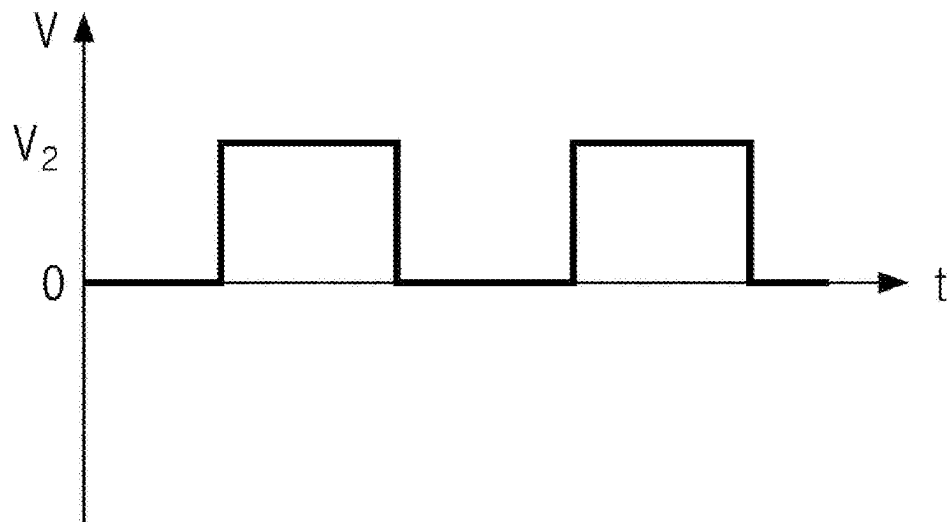
[FIG 1]



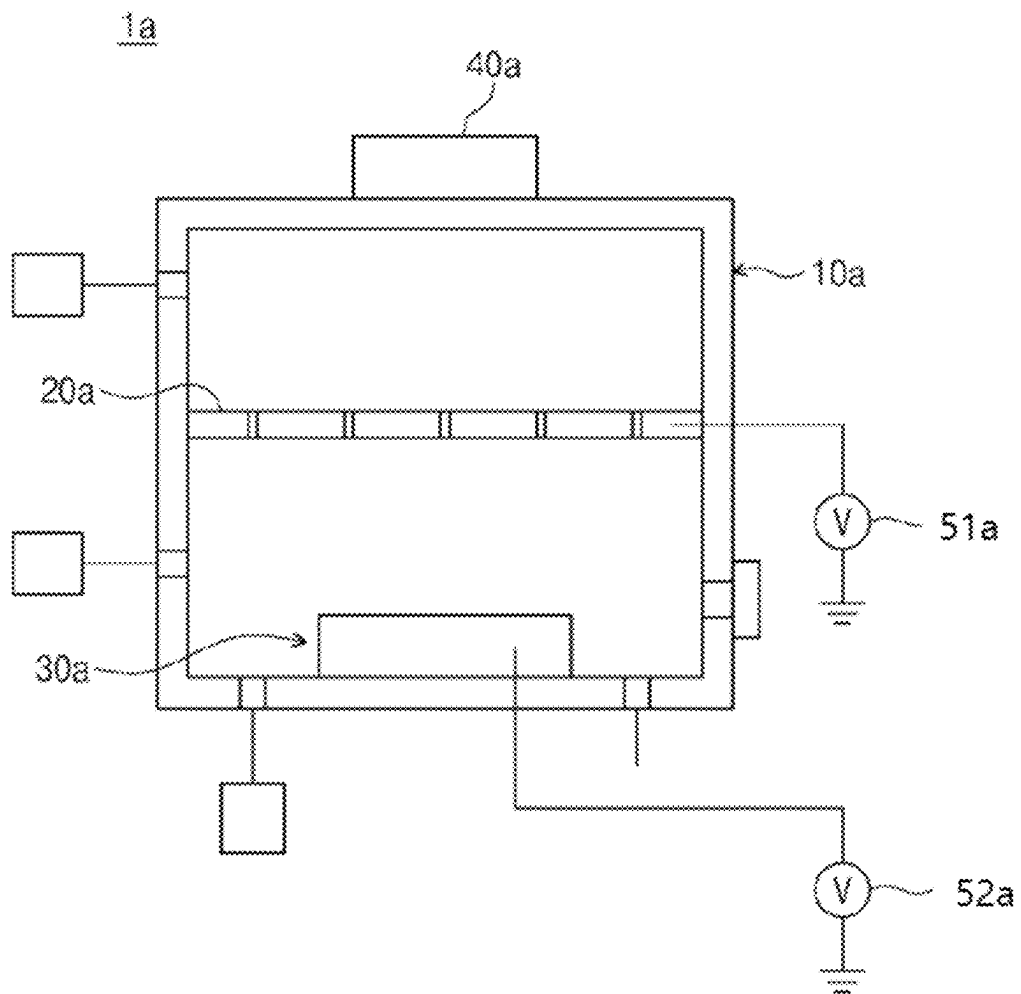
【 FIG 2】



【 FIG 3】



[FIG 4]



1

SUBSTRATE PROCESSING APPARATUS AND SUBSTRATE PROCESSING METHOD

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a National Stage of International Application No. PCT/KR2020/004441, filed Apr. 1, 2020, claiming priority to Korean Patent Application No. 10-2019-0068789, filed Jun. 11, 2019.

TECHNICAL FIELD

The present invention relates to substrate processing apparatus and a substrate processing method, and more particularly, to substrate processing apparatus and substrate processing method capable of processing a substrate in units of atomic thickness.

BACKGROUND ART

Plasma may be used in a process for processing a substrate. For example, plasma may be used for an etching process or a deposition process. Plasma is generated by a very high temperature, a strong electric field, or a high-frequency electromagnetic field (RF electromagnetic fields), and refers to an ionized gas state composed of ions, electrons, radicals, and the like. In a conventional etching process, ion particles contained in plasma collide with a substrate. In addition, a deposition process may be performed by supplying a deposition material to the upper portion of a substrate after the gas including a material to be deposited on the substrate reacts with the plasma.

Due to the energy of ions, in the process of processing a substrate by the ions, it may be processed or damaged in a shape different from the originally intended processing shape. As the line width of a device is miniaturized, such damage to the substrate degrades the device property.

DISCLOSURE

Technical Problem

An object of the present invention is to provide a substrate processing apparatus and a substrate processing method capable of processing a substrate in units of atomic thickness.

Another object of the present invention is to provide a substrate processing apparatus and a substrate processing method capable of processing a substrate using electrons.

Still another object of the present invention is to provide a substrate processing apparatus and a substrate processing method capable of processing a large-area substrate using electrons.

Still another object of the present invention is to provide a substrate processing apparatus and a substrate processing method capable of performing a process in a state in which damage to the substrate is minimized.

Technical Solution

According to an embodiment, there is provided a substrate processing apparatus that includes a chamber; a susceptor located inside the chamber to support a substrate to be processed; a first plasma control plate located in an upper area of a space inside the chamber; a second plasma control plate located below the first plasma control plate while being

2

spaced apart from the first plasma control plate by a predetermined distance; and a plasma control power source connected to the first plasma control plate and the second plasma control plate.

In addition, the plasma control power source may apply a voltage having a magnitude equal to or less than a ground state to the first plasma control plate.

In addition, the plasma control power source may apply a negative voltage in a form of a pulse to the first plasma control plate.

In addition, the plasma control power source may apply a voltage having a magnitude greater than or equal to the ground state to the second plasma control plate.

In addition, the plasma control power source may apply a positive voltage in a form of a pulse to the second plasma control plate.

In addition, the substrate processing apparatus may further include an upper gas supply source connected to a space formed above the first plasma control plate; and a lower gas supply source connected to a space formed below the second plasma control plate.

In addition, the upper gas supply source may supply a process gas reacting with the substrate to form a reaction layer, or an excitation gas excited by plasma.

In addition, the lower gas supply source may supply a process gas reacting with the substrate to form a reaction layer, or an excitation gas excited by plasma.

In addition, the plasma control power source may apply a voltage in a form of a pulse such that a sink of a voltage applied to the first plasma control plate and the second plasma control plate is adjustable.

According to another embodiment, there is provided a substrate processing apparatus that includes a chamber; a susceptor located inside the chamber to support a substrate to be processed; a plasma control plate located in a space inside the chamber; and a plasma control power source coupled to the plasma control plate and the susceptor.

In addition, the plasma control power source may apply a voltage having a magnitude equal to or less than a ground state to the plasma control plate and apply a voltage having a magnitude greater than or equal to the ground state to the susceptor.

According to still another embodiment, there is provided a substrate processing method that includes exciting plasma in an excitation space located above a first plasma control plate located in an upper area of an inner space of the chamber; moving electrons having energy adapted for overcoming a repulsive force caused by the first plasma control plate from the excitation space to a control space located below the first plasma control plate by applying a voltage having a magnitude equal to or less than a ground state to the first plasma control plate; and moving only electrons from the control space to a process space located below a second plasma control plate by applying a voltage having a magnitude greater than or equal to the ground state to the second plasma control plate located below the first plasma control plate.

In addition, the substrate processing method may further include reacting the process gas with a substrate located in the chamber to form a reaction layer after supplying a process gas into the chamber before exciting the plasma.

In addition, the reaction layer may be for an etching process, and the reaction layer reacts with electrons in plasma to be etched from the substrate.

In addition, the reaction layer may be for a deposition process, and the reaction layer reacts with a layer located under the reaction layer by energy provided from electrons in plasma.

Advantageous Effects

According to the embodiments of the present disclosure, it is possible to provide a substrate processing apparatus and a substrate processing method capable of processing a substrate in units of atomic thickness.

In addition, according to the embodiments of the present disclosure, it is possible to provide a substrate processing apparatus and a substrate processing method capable of processing a substrate using electrons.

In addition, according to the embodiments of the present disclosure, it is possible to provide a substrate processing apparatus and a substrate processing method capable of processing a large-area substrate using electrons.

In addition, according to the embodiments of the present disclosure, it is possible to provide a substrate processing apparatus and a substrate processing method capable of performing a process in a state in which damage to the substrate is minimized.

DESCRIPTION OF DRAWINGS

FIG. 1 is a view illustrating a substrate processing apparatus according to an embodiment of the present invention.

FIG. 2 is a diagram illustrating the voltage applied to the first plasma control plate according to an embodiment.

FIG. 3 is a diagram illustrating the voltage applied to the second plasma control plate according to an embodiment.

FIG. 4 is a view illustrating a substrate processing apparatus according to another embodiment of the present invention.

BEST MODE

Mode for Invention

Hereinafter, embodiments of the present invention will be described in detail with reference to accompanying drawings. Embodiments of the present invention may be modified in various forms, and the scope of the present invention should not be construed as being limited to the following embodiments. This embodiment is provided to more completely explain the present invention to those of ordinary skill in the art. Accordingly, the shapes of elements in the drawings are exaggerated to emphasize a clearer description.

FIG. 1 is a view illustrating a substrate processing apparatus according to an embodiment of the present invention.

Referring to FIG. 1, a substrate processing apparatus 1 includes a chamber 10, a plasma control plate 20, a susceptor 30, a plasma excitation unit 40, and a plasma control power source 50.

The substrate processing apparatus 1 processes a substrate by using plasma. For example, the substrate processing apparatus 1 may perform an etching process, a deposition process, an ashing process, and the like on a substrate by using excited plasma. In detail, the substrate processing apparatus 1 according to an embodiment of the present invention performs a process on a substrate in units of atomic layer.

The chamber 10 provides a space in which a substrate is placed and a process is performed. The chamber 10 may be

made of a conductive material such as aluminum or stainless steel. The chamber 10 may be provided in a grounded state.

The plasma control plate 20 is provided inside the chamber 10. The plasma control plate 20 is provided to have an area corresponding to an area along a cross-section of the inner space of the chamber 10 so that the inner space of the chamber 10 is partitioned in the vertical direction.

The plasma control plate 20 includes a first plasma control plate 21 and a second plasma control plate 22.

The first plasma control plate 21 is located in the upper area of the space inside the chamber 10. Holes are formed in the first plasma control plate 21. For example, holes may be formed in the first plasma control plate 21 in a grid shape. The first plasma control plate 21 may be made of a conductive material.

The second plasma control plate 22 is spaced apart from the first plasma control plate 21 by a set interval, and is positioned below the first plasma control plate 21. Thus, the space inside the chamber 10 is partitioned into an excitation space above the first plasma control plate 21, a control space between the first and second plasma control plates 21 and 22, and a process space below the second plasma control plate 22. Holes are formed in the second plasma control plate 22. As an example, holes may be formed in a grid shape in the second plasma control plate 22. The second plasma control plate 22 may be made of a conductive material.

An opening 101 through which a substrate may be loaded into or unloaded out of the process space is formed at one side of the chamber 10, and the opening 101 may be opened and closed by a gate valve 102.

The susceptor 30 is located inside the chamber 10 to support a substrate to be processed. The susceptor 30 is located in the process space.

An exhaust port 103 may be formed at a lower portion of the chamber 10. The exhaust port 103 may be connected to an exhaust member 104. The exhaust member 104 may provide a suction pressure to cause the exhaust to be made to the inner space of the chamber 10.

An upper gas supply hole 105 is formed in the chamber 10. An upper gas supply hole 105 may be formed in a sidewall of the chamber 10. The upper gas supply hole 105 is formed to be connected to the excitation space. The upper gas supply hole 105 is connected to an upper gas supply source 106. The upper gas supply source 106 supplies gas. For example, the gas supplied from the upper gas supply source 106 may be a compound containing an inert gas such as argon or helium, nitrogen, chlorine, or silicon.

A lower gas supply hole 107 may be formed in the chamber 10. The lower gas supply hole 107 may be formed in a sidewall of the chamber 10. The lower gas supply hole 107 is formed to be connected to the process space. The lower gas supply hole 107 is connected to a lower gas supply source 108. The lower gas source 108 supplies gas. The lower gas source 108 may be provided in the same or different configuration as the upper gas supply source 106. For example, the gas supplied from the lower gas supply source 108 may be provided as a compound containing an inert gas such as argon or helium, nitrogen, chlorine, or silicon.

A cover part 110 is positioned on the upper portion of the chamber 10, so that the inside of the chamber 10 is blocked. The cover part 110 may be provided in a plate shape to form an upper wall of the chamber 10. In addition, the cover part 110 may be provided in the shape of a container that is opened downward to form an upper wall and an upper side wall of the chamber 10.

The cover part **110** allows energy generated from the plasma excitation unit **40** to be transmitted into the inner space of the chamber **10**. The cover part **110** may be made of a dielectric material such as quartz. In addition, the cover part **110** may be a conductor having a slit formed therein.

The plasma excitation unit **40** is located above the chamber **10** and provides energy for excitation of the excitation gas supplied to the excitation space into plasma. For example, the plasma excitation unit **40** may be provided as an antenna positioned adjacent to the outer surface of the cover part **110** to provide energy for plasma excitation in an inductively coupled plasma scheme. In this case, the antenna may be provided to be positioned outside the upper wall of the chamber **10**, positioned outside the upper sidewall, or positioned outside the upper wall and the upper sidewall.

As another example, the plasma excitation unit **40** may be provided as a microwave application unit connected to the outer surface of the cover part **110**, so that the microwave generated from the microwave application unit penetrates the cover part **110** and then is supplied to the excitation space, thereby exciting the excitation gas into plasma.

The plasma control power source **50** controls the force applied to the plasma to control the state of the plasma used for processing the substrate, or the plasma state and the amount of energy. The plasma control power source **50** is connected to the first plasma control plate **21** and the second plasma control plate **22**.

FIG. 2 is a diagram illustrating the voltage applied to the first plasma control plate according to an embodiment. FIG. 3 is a diagram illustrating the voltage applied to the second plasma control plate according to an embodiment.

Referring to FIGS. 2 and 3, the plasma control power source **50** applies a voltage having a magnitude less than or equal to the ground state to the first plasma control plate **21**, and applies a voltage having a magnitude greater than or equal to the ground state to the second plasma control plate **22**.

The plasma control power source **50** may apply a negative voltage in the form of a pulse to the first plasma control plate **21**. In addition, the plasma control power source **50** may apply a voltage to the first plasma control plate **21** in such a way that a negative voltage of a predetermined magnitude is maintained for a predetermined time.

The plasma control power source **50** may apply a positive voltage in the form of a pulse to the second plasma control plate **22**. In addition, the plasma control power source **50** may continuously apply a positive voltage of a predetermined magnitude to the second plasma control plate **22** for a predetermined time.

A control unit (not shown) controls components of the substrate processing apparatus **1**.

Hereinafter, the operation of the substrate processing apparatus **1** will be described. As an example, a process in which an etching process is performed by the substrate processing apparatus **1** will be described.

The control unit causes the process gas to be supplied to the chamber **10**. The process gas may be supplied through the lower gas supply hole **107** or the upper gas supply source **106**. The process gas reacts with the upper surface of the substrate positioned on the susceptor **30** to form a reaction layer. In this case, the amount of the supplied process gas and the supply time of the process gas are controlled, so that the reaction layer may be formed to have an atomic thickness. For example, the process gas may be chlorine or the like.

The control unit controls the exhaust member **104** to exhaust the process gas from the chamber **10**.

Thereafter, the control unit causes the excitation gas to be supplied to the chamber **10**, and operates the plasma excitation unit **40** so that the excitation gas supplied to the chamber **10** is excited into a plasma state. Accordingly, positively charged ions and negatively charged electrons are generated in the excitation space. The excitation gas may be supplied through the upper gas supply source **106**. In addition, the excitation gas may be supplied through the lower gas supply source **108** to control the state of the process space in which the substrate is located.

The control unit operates the plasma control power source **50** together with or prior to the excitation of the plasma, thereby controlling a state in which the plasma generated in the excitation space is supplied to the process space. In detail, the first plasma control plate **21** has a negative voltage, so that a repulsive force is generated between the electrons and the first plasma control plate **21**. Accordingly, only electrons having energy capable of overcoming the repulsive force of the first plasma control plate **21** among electrons in the excitation space pass through the first plasma control plate **21** and are supplied to the control space. In addition, the amount of electrons supplied from the excitation space to the control space may be controlled by setting the voltage applied to the first plasma control plate **21** in a pulse form or by adjusting the magnitude of the negative voltage. In addition, since the second plasma control plate **22** has a positive voltage, a repulsive force is generated between the ions and the second plasma control plate **22**, so that the ions in the control space are blocked from moving into the process space. In addition, the electrons supplied into the control space receive a force in the process space direction by the repulsive force between the first plasma control plates **21** and the attractive force between the second plasma control plates **22**, so that the electrons are supplied to the process space in the state where energy is adjusted. Accordingly, the electrons react with the reaction layer of the substrate to perform etching at the atomic thickness. When a pulse type voltage is applied to the first and second plasma control plates **21** and **22**, the sink of the voltage applied to the first plasma control plate **21**, the second plasma control plate **22**, or the first and second plasma control plate **21** and **22** may be adjusted. For example, when the first plasma control plate **21** is $-V_1$, the second plasma control plate **21** may control the sink to be 0, or V_2 , or include a section in which the voltage changes from 0 to V or V_2 to 0.

Thereafter, the control unit controls the exhaust member **104** to discharge the etching by-products from the chamber **10**.

The above-described process may be performed once, or repeated twice or more, so that the process thickness may be adjusted by a multiple of the atomic thickness.

The substrate processing apparatus **1** according to an embodiment of the present invention performs the process processing by using electrons. In a conventional substrate process using ions, the reaction layer is etched by energy provided by ions (e.g., Ar^+ ions when argon is used as an excitation gas). In this case, excessive energy is applied to the substrate while the ions react with the substrate due to the mass of the ions, thereby causing damage to the substrate. Such damage causes the processed substrate to have a structure different from that originally intended, and increases the leakage current according to the miniaturization of the line width or increases the contact resistance in the metal deposition or annealing process, thereby reducing device performance. To the contrary, electrons have a very small mass compared to ions. Accordingly, the process of

processing a substrate using electrons by a substrate processing apparatus according to an embodiment of the present invention does not cause the above-described problem to the substrate during the processing process. In addition, the amount of electrons and energy of electrons used for processing the substrate are controlled by the plasma control power source 50, so that the process is performed with an atomic thickness while preventing the substrate from being damaged. In addition, since the process is performed by supplying the electrons extracted from the plasma to the substrate after excitation of the plasma in the excitation space, the process may be effectively performed even on a large-area substrate.

In addition, the substrate processing apparatus 1 may perform a deposition process.

The control unit causes the process gas to be supplied to the chamber 10. The process gas may be supplied through the lower gas supply hole 107 or the upper gas supply source 106. The process gas reacts with the upper surface of the substrate positioned on the susceptor 30 to form a reaction layer. For example, the process gas may be oxygen radical, a silicon source, or the like.

The control unit controls the exhaust member 104 to exhaust the process gas from the chamber 10.

Thereafter, the control unit causes the excitation gas to be supplied to the chamber 10, and operates the plasma excitation unit 40 so that the excitation gas supplied to the chamber 10 is excited into a plasma state. Thereafter, the control unit performs a control similar to that described above in the etching process, so that only electrons are selectively supplied to the substrate. Accordingly, the electrons supply energy for the reaction between the reaction layer and the material of the layer located below the reaction layer to react with each other.

Thereafter, the control unit controls the exhaust member 104 to discharge the process by-products from the chamber 10.

The above-described process may be performed once, or repeated twice or more, so that the thickness of the process may be adjusted by a multiple of the atomic thickness. In addition, materials of the sequentially deposited reaction layers may be the same or different from each other. For example, an oxygen layer is first formed on a substrate in atomic units, and a silicon layer is formed on an upper surface thereof in atomic units, so that a silicon oxide layer may be deposited at an atomic thickness.

The process gas and excitation gas in the above-described etching or deposition process are exemplary, and may be changed to those used as the process gas and excitation gas in the conventional etching process and deposition process.

Energy is required for a reaction to occur between the reaction layer and the substrate or between the reaction layers. In the case of a conventional substrate processing apparatus, such energy is supplied through a heater located in the susceptor or ions. Then energy provided by the heater or ions is relatively large and difficult to be controlled in small units, so that the energy is not suitable to be used when deposition is performed with an atomic thickness. To the contrary, the substrate processing apparatus 10 according to an embodiment of the present invention provides such energy through electrons, so that, when deposition is performed with an atomic thickness, a corresponding amount of energy may be effectively supplied.

FIG. 4 is a view illustrating a substrate processing apparatus according to another embodiment of the present invention.

Referring to FIG. 4, a substrate processing apparatus 1a includes a chamber 10a, a plasma control plate 20a, a susceptor 30a, a plasma excitation unit 40a, and a plasma control power source 50a.

The plasma control plate 20a is provided to have an area corresponding to an area along a cross-section of the inner space of the chamber 10a so that the inner space of the chamber 10a is partitioned in the vertical direction. Accordingly, an excitation space is formed above the plasma control plate 20a, and a process space is formed below the plasma control plate 20a. Holes are formed in the plasma control plate 20a. For example, holes may be formed in a grid shape in the plasma control plate 20a.

The plasma control power source 50a controls the force applied to the plasma to control the state of the plasma used for processing the substrate, or the plasma state and the amount of energy. The plasma control power source 50a is connected to the plasma control plate 20a and the susceptor 30a.

The plasma control power source 50a applies a voltage having a magnitude less than or equal to the ground state to the plasma control plate 20a, and applies a voltage having a magnitude greater than or equal to the ground state to the susceptor 30a.

The plasma control power source 50a may apply a negative voltage in the form of a pulse to the plasma control plate 20a. In addition, the plasma control power source 50a may apply a voltage to the plasma control plate 20 in such a way that a negative voltage of a predetermined magnitude is maintained for a predetermined time. Accordingly, when the control unit operates the plasma control power source 50a, the plasma control plate 20a has a negative voltage, so that a repulsive force is generated between the electrons and the plasma control plate 20a. Accordingly, only electrons having energy capable of overcoming the repulsive force caused by the plasma control plate 20a among the electrons in the excitation space pass through the plasma control plate 20a and are supplied to the process space.

In addition, the amount of electrons supplied from the excitation space to the process space may be controlled by setting the voltage applied to the plasma control plate 20a in a pulse form or by adjusting the magnitude of the negative voltage.

The plasma control power source 50 may apply a positive voltage in the form of a pulse to the susceptor 30a. In addition, the plasma control power source 50 may continuously apply a positive voltage of a predetermined magnitude to the susceptor 30a for a predetermined time. Accordingly, when the control unit operates the plasma control power source 50a, the susceptor 30a has a positive voltage, so that a repulsive force is generated between the ions and the susceptor 30a, thereby preventing the ions from moving toward the substrate. In addition, the electrons supplied to the process space receive a force toward the substrate by the repulsive force between the plasma control plates 20a and the attractive force between the susceptors 30a, so that the electrons react with the substrate in a state in which the state of energy is controlled.

Since the configuration of the chamber 10a, the susceptor 30a, the plasma excitation unit 40a, etc. in addition to the plasma control plate 20a and the plasma control power source 50a connected thereto in the substrate processing apparatus 1a according to the present embodiment is the same as or similar to the substrate processing apparatus of FIG. 1, the repeated description thereof will be omitted.

The process of processing a substrate by the substrate processing apparatus 1a according to the present embodiment

ment is similar to that of the substrate processing apparatus 1 of FIG. 1, and thus a repeated description will be omitted.

The above detailed description is illustrative of the present invention. In addition, the above description shows and describes preferred embodiments of the present invention, and the present invention may be used in various other combinations, modifications, and environments. That is, changes or modifications are possible within the scope of the concept of the invention disclosed herein, the scope equivalent to the written disclosure, and/or within the scope of skill or knowledge in the art. The written embodiments describe the best state for implementing the technical idea of the present invention, and various changes required in the specific application field and use of the present invention are possible. Therefore, the detailed description of the present invention is not intended to limit the present invention to the disclosed embodiments. In addition, the appended claims should be construed as including other embodiments.

The invention claimed is:

1. A substrate processing apparatus comprising:

- a chamber;
- a plasma excitation unit located on an upper part of the chamber;
- a susceptor located inside the chamber to support a substrate;
- a first plasma control plate located above the susceptor and forming an excitation space between the upper part of the chamber and thereof;
- a second plasma control plate located below the first plasma control plate and forming a control space between the first plasma control plate and thereof;
- a plasma control power supply applying voltage to the first plasma control plate and the second plasma control plate;

an upper gas supply source supplying an excitation gas to the excitation space;

a lower gas supply source supplying a process gas to a process space between the second plasma control plate and the susceptor; and

a control unit configured to:

control the lower gas supply source so that the process gas is supplied to the process space to form a reaction layer on the substrate,

control the upper gas supply source so that the excitation gas is supplied to the excitation space, and

control the plasma control power so that a negative voltage is applied to the first plasma control plate and a positive voltage is applied to the second plasma control plate to etch the reaction layer.

2. The substrate processing apparatus of claim 1, wherein each of the negative voltage and the positive voltage are a pulsed D.C. voltage.

3. The substrate processing apparatus of claim 1, wherein the control unit controls a first plasma control power source and a second plasma control power source so that when the negative voltage is applied to the first plasma control plate, the voltage applied to the second plasma control plate changes from 0 V to the positive voltage and from the positive voltage to 0 V.

4. The substrate processing apparatus of claim 1, wherein the process gas includes an inert gas, nitrogen, chlorine, and silicon.

5. The substrate processing apparatus of claim 1, wherein the reaction layer comprises:

- an oxygen layer formed on the substrate in atomic units; and
- a silicon layer formed on an upper surface of the oxygen layer in atomic units.

* * * * *